Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[11pt,a4paper]article [hyperref]acl2021 times latexsym

microtype

Instructions for ACL-IJCNLP 2021 Proceedings

First Author

Affiliation / Address line 1

Affiliation / Address line 2

Affiliation / Address line 3
email@domain
Second Author
Affiliation / Address line 1
Affiliation / Address line 2
Affiliation / Address line 3
email@domain

Ringkasan

This document contains the instructions for preparing a manuscript for the proceedings of ACL-IJCNLP 2021. The document itself conforms to its own specifications, and is therefore an example of what your manuscript should look like. These instructions should be used for both papers submitted for review and for final versions of accepted papers. Authors are asked to conform to all the directions reported in this document.

1 Credits

This document has been adapted by Roberto Navigli from the instructions for earlier ACL, NAACL and EMNLP proceedings, including those for EMNLP 2020 by Yulan He, ACL 2020 by Steven Bethard, Ryan Cotterrell and Rui Yan, ACL 2019 by Douwe Kiela and Ivan Vulić, NAACL 2019 by Stephanie Lukin and Alla Roskovskaya, ACL 2018 by Shay Cohen, Kevin Gimpel, and Wei Lu, NAACL 2018 by Margaret Michell and Stephanie Lukin, 2017/2018 (NA)ACL bibtex suggestions from Jason Eisner, ACL 2017 by Dan Gildea and Min-Yen Kan, NAACL 2017 by Margaret Mitchell, ACL 2012 by Maggie Li and Michael White, ACL 2010 by Jing-Shing Chang and Philipp Koehn, ACL 2008 by Johanna D. Moore, Simone Teufel, James Allan, and Sadaoki Furui, ACL 2005 by Hwee Tou Ng and Kemal Oflazer, ACL 2002 by Eugene Charniak and Dekang Lin, and earlier ACL and EACL formats written by several people, including John Chen, Henry S. Thompson and Donald Walker. Additional elements were taken from the formatting instructions of the *International Joint Conference on Artificial Intelligence* and the *Conference on Computer Vision and Pattern Recognition*.

2 Introduction

The following instructions are directed to authors of papers submitted to ACL-IJCNLP 2021 or accepted for publication in its proceedings. All authors are required to adhere to these specifications. Authors are required to provide a Portable Document Format (PDF) version of their papers. **The proceedings are designed for printing on A4 paper.**

3 Electronically-available resources

ACL provides this description and accompanying style files at

`https://2021.aclweb.org/files/acl-ijcnlp2021-templates.zip`

We strongly recommend the use of these style files, which have been appropriately tailored for the ACL-IJCNLP 2021 proceedings. You can also start from the Overleaf template linked from the conference website.

LaTeX-specific details: The templates include the LaTeX2e source (`acl2021.tex`), the LaTeX2e style file used to format it (`acl2021.sty`), an ACL bibliography style (`acl_natbib.bst`), an example bibliography (`acl2021.bib`), and the bibliography for the ACL Anthology (`anthology.bib`).

4 Length of Submission

The conference accepts submissions of long papers and short papers. Long papers may consist of up to eight (8) pages of content plus unlimited pages for references and ethics/broader impact statement. Short papers may consist of up to four (4) pages of content, plus unlimited pages for references and ethics/broader impact statement. Upon acceptance, final versions of both long and short papers will be given one additional content page to address the reviewers' comments.

For both long and short papers, all illustrations and tables that are part of the main text must be accommodated within these page limits, observing the formatting instructions given in the present document. **Papers that do not conform to the specified length and formatting requirements are subject to be rejected without review.**

The conference encourages the submission of additional material that is relevant to the reviewers but not an integral part of the paper. There are two such types of material: appendices, which can be read, and non-readable supplementary materials, often data or code. Additional material must be submitted as separate files, and must adhere to the same anonymity guidelines as the main paper. Appendices should be no longer than 4 pages. The paper must be self-contained: it is optional for reviewers to look at the supplementary material. The paper should not refer, for further detail, to documents, code or data resources that are not available to the reviewers. Refer to Section 8 for further information.

Workshop chairs may have different rules for allowed length and whether supplemental material is welcome. As always, the respective call for papers is the authoritative source.

5 Anonymity

As reviewing will be double-blind, papers submitted for review should not include any author information (such as names or affiliations). Furthermore, self-references that reveal the author's identity, *e.g.*,

> We previously showed Gusfield:97 ...

should be avoided. Instead, use citations such as

> Gusfield:97 previously showed...

Please do not use anonymous citations and do not include acknowledgements. **Papers that do not conform to these requirements may be rejected without review.**

Any preliminary non-archival versions of submitted papers should be listed in the submission form but not in the review version of the paper. Reviewers are generally aware that authors may

present preliminary versions of their work in other venues, but will not be provided the list of previous presentations from the submission form.

Please do not include github links that reveal the authors' identities in your submission. If you feel it is important to include your source code, you can zip the code and submit it to softconf.

Once a paper has been accepted to the conference, the camera-ready version of the paper should include the author's names and affiliations, and is allowed to use self-references and provide the related github link.

LaTeX-specific details: For an anonymized submission, ensure that `\aclfinalcopy` at the top of this document is commented out, and that you have filled in the paper ID number (assigned during the submission process on softconf) where `***` appears in the `\def\aclpaperid{***}` definition at the top of this document. For a camera-ready submission, ensure that `\aclfinalcopy` at the top of this document is not commented out.

6 Multiple Submission Policy

ACL-IJCNLP 2021 will not consider any paper that is under review in a journal or another conference at the time of submission, and submitted papers must not be submitted elsewhere during the ACL-IJCNLP 2021 review period. This policy covers all refereed and archival conferences and workshops (e.g., NAACL). The only exception is that a paper can be dual-submitted to both ACL-IJCNLP 2021 and an ACL-IJCNLP workshop.

In addition, we will not consider any paper that overlaps significantly in content or results with papers that will be (or have been) published elsewhere. Authors submitting more than one paper to ACL-IJCNLP 2021 must ensure that their submissions do not overlap significantly ($> 25\%$) with each other in content or results.

7 Formatting Instructions

Manuscripts must be in two-column format. Exceptions to the two-column format include the title, authors' names and complete addresses, which must be centered at the top of the first page, and any full-width figures or tables (see the guidelines in Section 7.5). **Type single-spaced.** Start all pages directly under the top margin. The manuscript should be printed single-sided and its length should not exceed the maximum page limit described in Section 4. Pages should be numbered in the version submitted for review, but **pages should not be numbered in the camera-ready version**.

LaTeX-specific details: The style files will generate page numbers when `\aclfinalcopy` is commented out, and remove them otherwise.

7.1 File Format

For the production of the electronic manuscript you must use Adobe's Portable Document Format (PDF). Please make sure that your PDF file includes all the necessary fonts (especially tree diagrams, symbols, and fonts with Asian characters). When you print or create the PDF file, there is usually an option in your printer setup to include none, all or just non-standard fonts. Please make

sure that you select the option of including ALL the fonts. **Before sending it, test your PDF by printing it from a computer different from the one where it was created.** Moreover, some word processors may generate very large PDF files, where each page is rendered as an image. Such images may reproduce poorly. In this case, try alternative ways to obtain the PDF. One way on some systems is to install a driver for a postscript printer, send your document to the printer specifying "Output to a file", then convert the file to PDF.

It is of utmost importance to specify the **A4 format** (21 cm x 29.7 cm) when formatting the paper. Print-outs of the PDF file on A4 paper should be identical to the hardcopy version. If you cannot meet the above requirements about the production of your electronic submission, please contact the publication chairs as soon as possible.

LaTeX-specific details: PDF files are usually produced from LaTeX using the `pdflatex` command. If your version of LaTeX produces Postscript files, `ps2pdf` or `dvipdf` can convert these to PDF. To ensure A4 format in LaTeX, use the command `\special{papersize=210mm,297mm}` in the LaTeX preamble (below the `\usepackage` commands) and use `dvipdf` and/or `pdflatex`; or specify `-t a4` when working with `dvips`.

7.2 Layout

Format manuscripts two columns to a page, in the manner these instructions are formatted. The exact dimensions for a page on A4 paper are:

- Left and right margins: 2.5 cm
- Top margin: 2.5 cm
- Bottom margin: 2.5 cm
- Column width: 7.7 cm
- Column height: 24.7 cm
- Gap between columns: 0.6 cm

Papers should not be submitted on any other paper size. If you cannot meet the above requirements about the production of your electronic submission, please contact the publication chairs above as soon as possible.

7.3 Fonts

For reasons of uniformity, Adobe's **Times Roman** font should be used. If Times Roman is unavailable, you may use Times New Roman or **Computer Modern Roman**.

Table 1 specifies what font sizes and styles must be used for each type of text in the manuscript.

LaTeX-specific details: To use Times Roman in LaTeX2e, put the following in the preamble:

```
\usepackage{times}
\usepackage{latexsym}
```

Type of Text	Font Size	Style
paper title	15 pt	bold
author names	12 pt	bold
author affiliation	12 pt	
the word "Abstract"	12 pt	bold
section titles	12 pt	bold
subsection titles	11 pt	bold
document text	11 pt	
captions	10 pt	
abstract text	10 pt	
bibliography	10 pt	
footnotes	9 pt	

Tabel 1: Font guide.

7.4 Ruler

A printed ruler (line numbers in the left and right margins of the article) should be presented in the version submitted for review, so that reviewers may comment on particular lines in the paper without circumlocution. The presence or absence of the ruler should not change the appearance of any other content on the page. The camera ready copy should not contain a ruler.

Reviewers: note that the ruler measurements may not align well with lines in the paper – this turns out to be very difficult to do well when the paper contains many figures and equations, and, when done, looks ugly. In most cases one would expect that the approximate location will be adequate, although you can also use fractional references (*e.g.*, this line ends at mark 295.5).

LaTeX-specific details: The style files will generate the ruler when `\aclfinalcopy` is commented out, and remove it otherwise.

7.5 Title and Authors

Center the title, author's name(s) and affiliation(s) across both columns. Do not use footnotes for affiliations. Place the title centered at the top of the first page, in a 15-point bold font. Long titles should be typed on two lines without a blank line intervening. Put the title 2.5 cm from the top of the page, followed by a blank line, then the author's names(s), and the affiliation on the following line. Do not use only initials for given names (middle initials are allowed). Do not format surnames in all capitals (*e.g.*, use "Mitchell" not "MITCHELL"). Do not format title and section headings in all capitals except for proper names (such as "BLEU") that are conventionally in all capitals. The affiliation should contain the author's complete address, and if possible, an electronic mail address.

The title, author names and addresses should be completely identical to those entered to the electronical paper submission website in order to maintain the consistency of author information among all publications of the conference. If they are different, the publication chairs may resolve the difference without consulting with you; so it is in your own interest to double-check that the information is consistent.

Start the body of the first page 7.5 cm from the top of the page. **Even in the anonymous version of the paper, you should maintain space for names and addresses so that they will fit in the final (accepted) version.**

7.6 Abstract

Use two-column format when you begin the abstract. Type the abstract at the beginning of the first column. The width of the abstract text should be smaller than the width of the columns for the text in the body of the paper by 0.6 cm on each side. Center the word **Abstract** in a 12 point bold font above the body of the abstract. The abstract should be a concise summary of the general thesis and conclusions of the paper. It should be no longer than 200 words. The abstract text should be in 10 point font.

7.7 Text

Begin typing the main body of the text immediately after the abstract, observing the two-column format as shown in the present document.

Indent 0.4 cm when starting a new paragraph.

7.8 Sections

Format section and subsection headings in the style shown on the present document. Use numbered sections (Arabic numerals) to facilitate cross references. Number subsections with the section number and the subsection number separated by a dot, in Arabic numerals.

7.9 Footnotes

Put footnotes at the bottom of the page and use 9 point font. They may be numbered or referred to by asterisks or other symbols.[1] Footnotes should be separated from the text by a line.[2]

7.10 Graphics

Place figures, tables, and photographs in the paper near where they are first discussed, rather than at the end, if possible. Wide illustrations may run across both columns. Color is allowed, but adhere to Section 9's guidelines on accessibility.

Captions: Provide a caption for every illustration; number each one sequentially in the form: "Figure 1. Caption of the Figure." "Table 1. Caption of the Table." Type the captions of the figures and tables below the body, using 10 point text. Captions should be placed below illustrations. Captions that are one line are centered (see Table 1). Captions longer than one line are left-aligned (see Table 2).

LaTeX-specific details: The style files are compatible with the caption and subcaption packages; do not add optional arguments. **Do not override the default caption sizes.**

[1] This is how a footnote should appear.

[2] Note the line separating the footnotes from the text.

Command	Output	Command	Output
{\"a}	ä	{\c c}	ç
{\^e}	ê	{\u g}	ğ
{\‘i}	ì	{\l}	ł
{\.I}	İ	{\~n}	ñ
{\o}	ø	{\H o}	ő
{\’u}	ú	{\v r}	ř
{\aa}	å	{\ss}	ß

Tabel 2: Example commands for accented characters, to be used in, *e.g.*, BibTeX names.

Output	natbib command	Old ACL-style command
Gusfield:97	\citep	\cite
Gusfield:97	\citealp	no equivalent
Gusfield:97	\citet	\newcite
Gusfield:97	\citeyearpar	\shortcite

Tabel 3: Citation commands supported by the style file. The style is based on the natbib package and supports all natbib citation commands. It also supports commands defined in previous ACL style files for compatibility.

7.11 Hyperlinks

Within-document and external hyperlinks are indicated with Dark Blue text, Color Hex #000099.

7.12 Citations

Citations within the text appear in parentheses as Gusfield:97 or, if the author's name appears in the text itself, as Gusfield:97. Append lowercase letters to the year in cases of ambiguities. Treat double authors as in Aho:72, but write as in Chandra:81 when more than two authors are involved. Collapse multiple citations as in Gusfield:97,Aho:72.

Refrain from using full citations as sentence constituents. Instead of

> "Gusfield:97 showed that ..."

write

> "Gusfield:97 showed that ..."

LaTeX-specific details: Table 3 shows the syntax supported by the style files. We encourage you to use the natbib styles. You can use the command `\citet` (cite in text) to get "author (year)" citations as in Gusfield:97. You can use the command `\citep` (cite in parentheses) to get "(author, year)" citations as in Gusfield:97. You can use the command `\citealp` (alternative cite without parentheses) to get "author year" citations (which is useful for using citations within parentheses, as in Gusfield:97).

7.13 References

Gather the full set of references together under the heading **References**; place the section before any Appendices. Arrange the references alphabetically by first author, rather than by order of occurrence in the text.

Provide as complete a citation as possible, using a consistent format, such as the one for *Computational Linguistics* or the one in the *Publication Manual of the American Psychological Association* APA:83. Use full names for authors, not just initials.

Submissions should accurately reference prior and related work, including code and data. If a piece of prior work appeared in multiple venues, the version that appeared in a refereed, archival venue should be referenced. If multiple versions of a piece of prior work exist, the one used by the authors should be referenced. Authors should not rely on automated citation indices to provide accurate references for prior and related work.

The following text cites various types of articles so that the references section of the present document will include them.

- Example article in journal: Ando2005.
- Example article in proceedings, with location: borschinger-johnson-2011-particle.
- Example article in proceedings, without location: andrew2007scalable.
- Example arxiv paper: rasooli-tetrault-2015.

LaTeX-specific details: The LaTeX and BibTeX style files provided roughly follow the American Psychological Association format. If your own bib file is named `acl2021.bib`, then placing the following before any appendices in your LaTeX file will generate the references section for you:

```
\bibliographystyle{acl_natbib}
\bibliography{acl2021}
```

You can obtain the complete ACL Anthology as a BibTeX file from `https://aclweb.org/anthology/anthology.bib.gz`. To include both the anthology and your own bib file, use the following instead of the above.

```
\bibliographystyle{acl_natbib}
\bibliography{anthology,acl2021}
```

7.14 Digital Object Identifiers

As part of our work to make ACL materials more widely used and cited outside of our discipline, ACL has registered as a CrossRef member, as a registrant of Digital Object Identifiers (DOIs), the standard for registering permanent URNs for referencing scholarly materials.

All camera-ready references are required to contain the appropriate DOIs (or, as a second resort, the hyperlinked ACL Anthology Identifier) to all cited works. Appropriate records should be found for most materials in the current ACL Anthology at `http://aclanthology.info/`. As examples, we cite goodman-etal-2016-noise to show you how papers with a DOI will appear in the bibliography. We cite harper-2014-learning to show how papers without a DOI but with an ACL Anthology Identifier will appear in the bibliography.

LaTeX-specific details: Please ensure that you use BibTeX records that contain DOI or URLs for any of the ACL materials that you reference. If the BibTeX file contains DOI fields, the paper title in the references section will appear as a hyperlink to the DOI, using the hyperref LaTeX package.

8 Supplementary Materials

Supplementary material may report preprocessing decisions, model parameters, and other details necessary for the replication of the experiments reported in the paper. Seemingly small preprocessing decisions can sometimes make a large difference in performance, so it is crucial to record such decisions to precisely characterize state-of-the-art methods.

Nonetheless, supplementary material should be supplementary (rather than central) to the paper. **Submissions that misuse the supplementary material may be rejected without review.** Supplementary material may include explanations or details of proofs or derivations that do not fit into the paper, lists of features or feature templates, sample inputs and outputs for a system, hyperparameter values, pseudo-code or source code, and data.

The paper should not rely on the supplementary material: while the paper may refer to and cite the supplementary material and the supplementary material will be available to the reviewers, they will not be asked to review the supplementary material.

Supplementary material can be uploaded as up to three separate files: a single appendix file in PDF format, a single .tgz or .zip archive containing software, and/or a single .tgz or .zip archive containing data.

8.1 Appendices

Appendices are material that can be read, and include lemmas, formulas, proofs, and tables that are not critical to the reading and understanding of the paper. In the submitted version, appendices should be uploaded as a separate file. **Submissions which include appendices in the main paper will be rejected without review.** In the camera ready version, instead, appendices should directly follow the text and the references. In all cases, appendices should use the same format as the main paper.

LaTeX-specific details: In your camera ready version use `\appendix` before any appendix section to switch the section numbering over to letters.

8.2 Software and Data

Submissions may include software and data used in the work and described in the paper. Any accompanying software and/or data should include licenses and documentation of research review as appropriate.

9 Accessibility

In an effort to accommodate people who are color-blind (as well as those printing to paper), grayscale readability is strongly encouraged. Color is not forbidden, but authors should ensure that tables

and figures do not rely solely on color to convey critical distinctions. A simple criterion: All curves and points in your figures should be clearly distinguishable without color.

10 Translation of non-English Terms

It is also advised to supplement non-English characters and terms with appropriate transliterations and/or translations since not all readers understand all such characters and terms. Inline transliteration or translation can be represented in the order of:

original-form
transliteration
"translation"

11 LaTeX Compilation Issues

You may encounter the following error during compilation:

> `\pdfendlink` ended up in different nesting level than `\pdfstartlink`.

This happens when `pdflatex` is used and a citation splits across a page boundary. To fix this, the style file contains a patch consisting of two lines: (1) `\RequirePackage{etoolbox}` (line 455 in `acl2021.sty`), and (2) A long line below (line 456 in `acl2021.sty`).

If you still encounter compilation issues even with the patch enabled, disable the patch by commenting the two lines, and then disable the `hyperref` package by loading the style file with the `nohyperref` option:

```
\usepackage[nohyperref]{acl2021}
```

Then recompile, find the problematic citation, and rewrite the sentence containing the citation. (See, *e.g.*, `http://tug.org/errors.html`)

Acknowledgments

The acknowledgments should go immediately before the references. Do not number the acknowledgments section. **Do not include this section when submitting your paper for review.**